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The Editor-in-Chief
Biomedical Signal Processing and Control

Dear Editor,

Please consider the enclosed manuscript, "Leakage-controlled evaluation of stacking and recursive feature elimination for diabetes outcome prediction on the PIMA dataset", for publication as a Research Article in *Biomedical Signal Processing and Control*.

The manuscript addresses the processing, analysis and evaluation of physiological measurements for computer-aided decision making in medicine. Its predictors are physiological measurements: post-load plasma glucose and insulin from an oral glucose tolerance test, blood pressure and anthropometric measures. The pipeline it studies is a standard biomedical data-analysis chain of preprocessing, missing-value handling, feature construction, wrapper feature selection and ensemble classification. We believe this methodological contribution is relevant to the journal's readers who develop and evaluate data-driven decision systems from biomedical measurements.

Stacking ensembles and wrapper feature selection are widely reported for diabetes prediction, often with very high accuracy on the PIMA Indians Diabetes dataset. Such results are hard to interpret when imputation or feature selection is fitted before validation and when differences between models are not tested statistically. Our study takes a methodological rather than an algorithmic approach:

- every data-dependent step (imputation, feature engineering, scaling, RFECV and stacking) is re-fitted inside each validation split;
- eleven learners (the eight base learners of the stack, two reference learners and the stack itself) are compared with and without a common RFECV subset using $5\times10$-fold repeated cross-validation, corrected confidence intervals and multiplicity-adjusted tests, and confirmed on an untouched hold-out set with DeLong tests and bootstrap intervals;
- the whole framework is replicated in an independent, population-based NHANES 2013–2014 cohort with an HbA1c-defined outcome;
- the evaluation goes beyond discrimination to calibration, sensitivity-targeted operating thresholds, decision curves, a missing-data sensitivity analysis and exact SHAP attributions.

The main finding is deliberately reported without overstatement. RFECV produces compact feature sets with a stable core of predictors, and the stacking ensemble gives the highest observed hold-out sensitivity and MCC. However, it does not discriminate better than regularised logistic regression once validation is leakage-free, in either cohort, and its class-weighted probabilities are systematically too high, a problem that accuracy-based reporting would hide. Because the PIMA predictors include 2-h OGTT values, we present the work as a methodological benchmark of internal validity, not as a deployable screening tool. We believe these results, together with fully reproducible code, will be useful to readers who develop and review biomedical classification systems.

This manuscript is original, has not been published previously and is not under consideration by any other journal. The author has no competing interests to declare. All code and results are

publicly available.

Yours sincerely,

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